The Waterfront Brief

Weekly sustainment intelligence for naval shipyard leadership · Naval shipbuilding, ship repair, and fleet sustainment

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Five stories this week, and the through line is how long the answers take. The submarine chief has locked the Virginia-class design and is moving fabrication out of the two building yards, aimed at fiscal 2031. The Government Accountability Office has put the rebuild of the four public shipyards past $200 billion and past 2080, with the shops queued behind the docks. The first Navy warfare center to hold a NAVSEA-approved metal printing procedure has one good for a single steel, after seven years, and is offering it to other commands to copy. A third homeport has its sailor-led repair command back, with no shops and no billet count published. And the foreign-build plans due in November are still paper, so the part to watch is the trades they would have to hire, in a labor market the record says was cut and has not come back.

How to read this: where a sentence names where something came from, those words are a link to the source itself, and the sentence says what that source says. A sentence with no link is this desk's analysis, built only on the linked facts above it. Two warnings are still written out in the text: Speculation, and Unverified for a claim nobody here could check against a primary source, with the reason it could not be checked.

> **Correction, 2026-10-24:** The Mayport SIMA Assessment now gives GAO's September 2024 count; GAO has since closed the single-owner recommendation. Ruled by the editor; logged on the corrections page.
> **Correction, 2026-10-24:** The 38,845 census figure covers all 13 federal ship and boat building sites, not only the four public yards, and is now labeled so. Ruled by the editor; logged on the corrections page.

WATERFRONT OPS & PRODUCTION

Navy freezes Virginia-class design changes and moves more submarine work outside the two prime yards

The Navy's submarine chief has locked the Virginia-class design and is moving millions of man-hours a year of structural fabrication, machining and electrical work out of the two building yards.

Naval News reported on September 21 that Vice Adm. Rob Gaucher has ordered "pencils down" on Virginia-class design changes unless safety or a critical operational need requires it, with his personal sign-off required for an exception. Gaucher is dual-titled as Direct Reporting Portfolio Manager for Submarines. Deck-plate adjustments "such as moving a pipe hanger to ensure fit" stay delegated.

The Navy's May 2026 shipbuilding plan states that manager's goal as "a production rate of at least one Columbia SSBN and two Virginia SSNs per year by FY31". An SSBN is a ballistic-missile submarine and an SSN an attack submarine. Fiscal 2031 begins in October 2030.

The same Naval News report says the two prime yards, General Dynamics Electric Boat and HII Newport News Shipbuilding, keep module outfitting, hull assembly and final testing, while component assembly moves outside "so that major modules arrive fully outfitted". It names Rhoads, Austal and BAE Systems as "strategic outsourcers", firms outside the two nuclear shipbuilders that take their fabrication work.

Four Navy witnesses told a Senate Seapower subcommittee in April 2025 that the class reached "a build rate close to 1.9 per year" before the pandemic. It then dropped "to a production rate of 1.13 at the end of CY 2024". Of the four drivers they name, one is "lead ship issues associated with the Virginia Payload Module variant".

The House Appropriations Committee reported on June 26 that "Virginia-class production remains near 1.1 boats per year", and that the program is "delayed by as much as 42 months". Its bill orders quarterly Navy reporting of "the realized construction cadence against the cadence of record" and "the workforce required by trade and fiscal year".

GAO, Congress's audit agency, reported in July 2026 that the Navy took delivery of two Block IV submarines in 2025 and "both were over 3 years late". It says the program's own progress metrics "indicate the program is already underperforming its recovery plans".

U.S. Naval Institute News reported on April 21 that Rear Adm. Todd Weeks, under Gaucher, owns in-service submarines.

Gaucher gave the hours per hull in March, saying it takes "about 13 million hours to build a basic Virginia-class submarine and 18 million for one with payloads", and that "a Columbia-class submarine takes 34 million hours to make". Adding those up, he said, gives "about 70 million hours that we need to generate to get to that two-plus-one case", meaning two attack boats and one missile boat a year.

The Congressional Research Service quotes the Navy's own workload arithmetic: a Columbia counts as "approximately 2.5" Virginias "in terms of build resources (manning) and tonnage", and a payload-module boat as 1.25. So "a 1.0 build rate from 2015 becomes 5.0 in 2028". GAO reported in July 2026 that the payload module "makes the submarines 30 percent larger".

U.S. Naval Institute News reported on May 12 that the Chief of Naval Operations told House appropriators the two-a-year rate would arrive "around 2032".

The most repeated number here is not what it looks like. The 70 million hours is a year's work for three hulls, not one, and the five is the Navy's own workload factor for going from one boat a year to two plus a missile boat. Per hull the growth is 13 million to 18 million on a boat GAO calls 30 percent larger, which is close to proportionate. What has grown is duration, not difficulty. And production stood at 1.13 at the end of 2024 against near 1.1 now, so on the Navy's own measure it has not moved while the target went up. A design freeze reaches one of the four drivers named, and that one is a lead-ship problem on the first Payload Module hull rather than churn across the class. Expect fiscal 2031 to be missed and the rate to land nearer the 2032 the Chief of Naval Operations gave.

WHY IT MATTERS

Bottom line: treat fiscal 2031 as a date to check, not one to plan around. Ask whether the sign-off rule reaches alteration packages on hulls already in service, which sit under Weeks and which no source read here settles. Then watch for the first quarterly cadence report Congress has ordered, which will carry the realized rate and the workforce by trade.

Sources: Naval News, Sep 21, 2026; U.S. Navy Shipbuilding Plan, May 2026; U.S. Naval Institute News, May 12, 2026; U.S. Naval Institute News, Apr 21, 2026; Senate Armed Services Seapower subcommittee hearing, Apr 8, 2025 (S. Hrg. 119-269); H. Rept. 119-715, Jun 26, 2026; Defense One, Mar 20, 2026; Congressional Research Service RL32418, Jan 26, 2026; GAO-26-108457, Jul 2, 2026; Vol. 1 No. 10 (week 38)

The shipyard rebuild now runs past 2080, and the shops come last

This brief's first issue said to watch whether the money for the public shipyard rebuild held. The Government Accountability Office has now priced it, and published the order the work is queued in.

GAO reported on September 25 that the Navy's Shipyard Infrastructure Optimization Program "will surpass $200 billion and take over 50 more years to execute, based on GAO's analysis of Navy plans". The program began in 2018 and rebuilds the four public naval shipyards.

The same report says the Navy initially estimated the program "would cost at least $21 billion over 20 years".

GAO's report says the Navy issued guidance in 2025 setting annual funding expectations and directing that the plans "and the number and type of planned projects to be scheduled every year, be updated to reflect this guidance". The program now runs in three blocks. Block 1, roughly fiscal 2026 through 2042, is dry docks and piers plus "the first new waterfront production facility at each shipyard". Block 2, roughly 2038 through 2079, is "all the remaining waterfront production facilities". Block 3, from about 2074, is "logistics complexes, specialized repair shops, administrative support facilities". GAO says the last planned project finishes after 2080, and that officials put new facilities for some maintenance functions "into the 2070s or beyond".

The same report says the Ford and Virginia Block V classes are "not yet fully supported by existing public shipyard dry-dock capacity".

Per the same report, concurrent construction and shipyard operations "create challenges to ensuring the shipyards can complete their ship maintenance requirements", and all four yards report the same trouble. At Puget Sound the dry-dock project demolishes a large maintenance facility next to Dry Dock 3 and moves its occupants off the yard, and the utility upgrades behind it "require electricity outages that can cause major disruptions".

The same report says the program had bought 385 pieces of equipment, cranes among them, worth $1.49 billion by March 2026, on procurement money rather than construction money.

Read as a sequence, that plan moves the binding constraint off the dock and onto the shop. Docking capacity today does not fully support the two classes the yards must take, and the block ending around 2042 is where that gets fixed. In that same period each yard gets one new production building, and the rest wait on a block running to 2079. Because the plans and the yearly project count were rewritten to match a funding figure rather than the other way round, expect the shop side to absorb the slip each time a dock project grows. The equipment line is the exception: $1.49 billion of machinery has moved with no building attached, so a shop can get newer tools long before a newer roof.

WHY IT MATTERS

Bottom line: plan for new docks and old shops for the rest of your working life, and cost work around construction rather than after it. Ask which block the facility your trade depends on sits in, and get your yard's demolition and outage sequence in writing before you commit to a date. Watch whether Congress takes GAO's suggestion of an annual status report, because without one a block slipping is visible only from inside.

Sources: GAO-26-107830, Sep 25, 2026; Vol. 1 No. 1 (week 29)

MAINTENANCE ENGINEERING & TECHNOLOGIES

Keyport has NAVSEA approval to print one steel, and the alloy it calls more useful is still in qualification

A Navy engineering command now holds a NAVSEA-approved procedure for printing metal parts, and says other government-run sites can copy it.

Naval Undersea Warfare Center Division, Keyport said in an August 5 release that it is "the first Warfare Center to receive official approval from Naval Sea Systems Command for a metal powder bed fusion additive manufacturing procedure". A Warfare Center is a NAVSEA engineering and test command, not a shipyard.

Per the same release, the approval covers producing "tactical- and shipboard-level parts using 17-4PH stainless steel" on an M290 printer from EOS, a commercial maker of laser powder bed machines. It removes "the need for repeated baseline and machine qualifications for future 17-4PH prints". In powder bed fusion a laser melts fine metal powder into a solid part, layer by layer.

The release says the effort began seven years ago as a funded research project and "later evolved into a formal qualification project with support from the Submarine

Industrial Base". That is the Navy's investment program for submarine build and repair capacity.

The release quotes Bryce Weber, a Keyport technology lead, saying the "processes to achieve this qualification can be transferred to other commands and duplicated". He says that gives other government-run sites an easier "instruction set" and more qualified machines able to print critical shipboard parts during availabilities.

The release also says the command is "working on qualifying the printer for 316L stainless steel", an alloy it calls broader in application for ship and submarine repair.

Take the release at its word and the arithmetic is thin. Seven years bought one alloy on one machine model, and the command itself names the alloy it has not finished as the more useful one for ship and submarine repair. That makes the transfer claim the only part with schedule value, and it is testable: a second command approved inside a year shows the procedure travels, while another multi-year campaign means every site pays the full price again. Expect nothing in a repair package to move until 316L clears.

> WHY IT MATTERS
> **Bottom line:** do not plan a printed part into a package on the strength of this approval. Ask NAVSEA 05T's additive manufacturing technical warrant holder whether a duplicated procedure inherits Keyport's qualification or owes its own. Ask Keyport when it expects 316L. Then watch for the second command approved under the same procedure, because that date says whether printed metal reaches a waterfront.

Sources: NUWC Division, Keyport via DVIDS, Aug 5, 2026; Vol. 1 No. 7 (week 35)

OPERATIONS MAINTENANCE & READINESS

Mayport is the third homeport to get its sailor-led repair command back

The Navy reopened a Shore Intermediate Maintenance Activity, or SIMA, at Naval Station Mayport on September 15, the third this year. A SIMA does intermediate-level repair, the tier between what a crew does alongside and what a shipyard does with the ship open, so it decides which small jobs finish at the homeport rather than in a yard package.

The Navy said in a release on September 16 that SIMA Mayport was reestablished the day before at Southeast Regional Maintenance Center, and that SIMA Norfolk and SIMA San Diego were reestablished on June 2 and June 4. It names no shops, no billets and no ships. A Regional Maintenance Center is the Navy's standing shore repair organization for a homeport region.

This brief has run the thread three times. What a third opening adds is who owns the result.

GAO reported in September 2024 that 63 percent of the executive officers in its fleet-wide survey found it moderately to extremely difficult to complete repairs under way with the sailors assigned to their ships. It also found the Navy's step-by-step guidelines for shipboard jobs "sometimes inaccurate with respect to the time and personnel needed".

GAO reported in January 2025 that seven recommendations from that report were still not implemented as of September 2024. The Navy had also still not designated "a single entity ... to address challenges affecting intermediate maintenance periods", asked for in 2022.

The two SIMA commanding officers said on September 3 that before June, SIMA San Diego and SIMA Norfolk "previously operated as production departments under Regional Maintenance Centers". They said both commands are "establishing additional officer billets", and that SIMA has "assumed responsibility for Sailors assigned to Fleet Technical Assist". The Navy has said none of this about Mayport, and no billet count is published for any of the three.

The Navy's fiscal 2027 operations and maintenance justification asks a $67.5 million increase for intermediate-level maintenance at the Southeast, Mid-Atlantic and Forward Deployed regional maintenance centers. It adds 545 civilian full-time equivalent positions, which are funded slots rather than named people. The book never uses the word SIMA, so those are center positions rather than the new commands' own.

On the published record these are not new organizations. Two of the three were production departments inside their parent centers, broken out as commands, with the technical assist sailors folded in and some officer billets added. The people were already there. So the payoff the Navy claims sits on the ship rather than in the new command's shop, and the measure that matters is whether a crew can finish a job under way. The last public reading was 63 percent of executive officers calling that moderately to extremely difficult, against guidelines GAO found understate the people needed. GAO's September 2024 count, the one this article relied on, had seven recommendations still open and no single owner designated for these periods. Expect no published figure through fiscal 2027 showing intermediate-level work at Mayport taken back from contractors.

> WHY IT MATTERS
> **Bottom line:** coordination is not arbitration, and nothing published says who decides. Ask Southeast Regional Maintenance Center and SIMA Mayport which job types moved into the new command's shops, and who breaks the tie when both want the same technician. Then plan against the shops you can name.

Sources: Navy via DVIDS, Sep 16, 2026; SIMA Norfolk via DVIDS, Jun 2, 2026; Navy via DVIDS, Jun 5, 2026; U.S. Navy on SIMA San Diego, Jun 5, 2026; GAO-24-106525, Sep 2024; GAO-25-106990, Jan 2025

BUSINESS PRACTICES & STRATEGY

The leadership column does not run this week. Nothing in the pipeline had a study behind it, and the slot is earned rather than filled.

The foreign-build plans due in November have to find their trades in towns that are already short

Nothing has been ordered built. What is due in November is paperwork: two plans for how the Navy might buy ships designed abroad and built here, plus a study of which designs. No yard has been chosen and no supplier has agreed to anything. What is worth watching is that any answer puts another buyer into a trades market that is already short.

The President's memorandum of August 13 determines that building to a foreign design is in the national security interest and orders the Secretary of War to submit two plans within 90 days, which falls on November 11. One covers surface combatants, the other replenishment tankers and roll-on, roll-off vessels. It sets terms of entry rather than commitments anyone has made. A qualifying foreign supplier buys or builds an American yard, "hires and trains an American citizen workforce for the United States shipyards", licenses its parent yard's techniques, and sources an American supply chain "for construction and maintenance of all ships". No contract may be signed until 30 days after the Secretary notifies Congress.

U.S. Naval Institute News reported on September 1 that the acting Navy Secretary separately directed a study due November 12, covering combatants plus roll-on, roll-off ships and replenishment tankers for Military Sealift Command.

The Navy's Maritime Industrial Base program reported in January that its recruiting site drew "more than 435,000 job applications to MIB companies (107 percent increase)" during 2025. Interest is not the shortage.

The Bureau of Labor Statistics wage file for May 2025 shows the shortage is priced by trade. Shipbuilding pays its welders and machinists about four thousand dollars a year more than those trades average across all industries. It pays its pipefitters and electricians nine to eleven thousand a year less. A shipbuilding pipefitter's annual mean is $61,130 against $72,170 for the trade nationally.

GAO reported in February 2025 that five of the six Norfolk ship repair companies it spoke to had laid people off or feared they would, "because of a downturn in Navy repair work". The Navy meanwhile "projects that Norfolk will have either excess workforce capacity or the ability to provide surge capacity through overtime through at least fiscal year 2031".

The same program's metropolitan file puts a pipefitter's mean hourly wage in that metro at $28.50, below its $30.83 mean for every occupation of any kind, with 5,100 of them working there.

The Bureau's quarterly census puts 2025 federal ship and boat building employment, the four public naval shipyards and the Navy's other federal sites, at 38,845, below the 39,156 it recorded for 2019. Private ship and boat building fell in 2025 too.

The private yards have started paying up, and it moves people. GAO reported in April 2026 that after wage rises at a yard the Navy helped fund, a nearby shipbuilder said that "in a 2-week period they lost several workers to the shipyard that increased wages".

The question is not which design wins. It is that people are applying and the work was not steady. In the largest repair market in the country the firms were shedding trades for want of Navy work, while the Navy's own projection said that town had workers to spare. It is the same town where a pipefitter earns less than the average for every job in the metro. A dominant buyer in a surplus market has no reason to raise the price, and it did not. So whatever the November plans propose, a new yard is a new buyer in whatever town it lands, hiring the same people away from whoever holds them now. GAO has watched that happen in two weeks.

> **WHY IT MATTERS**
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> **Bottom line:** the plans are paper, so do not size anything to them. Size your trades against the split instead, because welders and machinists are paid competitively for this work and pipefitters and electricians are not, and that is where you lose people first. When the plans land, read them for the metro before the design, because that decides who it hires. The Navy's own regional officials say private industry can grow a workforce given stable work, and stability is the half nobody has delivered.

Sources: The White House, Aug 13, 2026; USNI News, Sep 1, 2026; Navy Maritime Industrial Base, Year in Review 2025, Jan 7, 2026; BLS Occupational Employment and Wage Statistics, May 2025; BLS Quarterly Census of Employment and Wages, 2025; GAO-25-106286, Feb 27, 2025; GAO-26-109068, Apr 22, 2026; HII, Mar 12, 2026; Vol. 1 No. 6 (week 34)

The Waterfront Brief is compiled weekly by the B2 vault's research engine from primary sources and verified trade reporting. Scope: US shipbuilding, US Navy sustainment, and partners across the Indo-Pacific region; out-of-area items only when there is a direct, stated lesson. Freshness policy: every article's anchor source is under 12 months old; older material appears only as dated context or a flagged exception. A linked sentence means the named source says it, not that this desk has independently verified it. Corrections welcome; staged sources are retained for audit.

UPCOMING EVENTS

Nearest first, filling the page. **Bold** marks a decision that closes inside the next 60 days. ASNE is the American Society of Naval Engineers, and NSRP is the National Shipbuilding Research Program, the industry-Navy body that funds shared shipyard process research. NSL is the Naval Submarine League, SNA is the Surface Navy Association, and USNI is the United States Naval Institute, which runs WEST with AFCEA, the Armed Forces Communications and Electronics Association. AMPP is the Association for Materials Protection and Performance, the corrosion and coatings society. DSI is the Defense Strategies Institute. SAE is SAE International, the engineering standards body that runs the Department of War symposium, and the AMDA Foundation is the Australian body behind the Indo Pacific exposition.

Dates	Event	Register by	Why it matters
Oct 20-21, 2026	NSRP Workforce & Compliance and Production Planning Joint Panel, Pensacola, FL (NSRP)	**In-person closes Oct 8**	Workforce and production-planning panels, jointly; virtual option
Nov 12-13, 2026	Naval Submarine League Annual Symposium 2026, Arlington, VA (NSL)	To be posted	Submarine industrial base: the submarine force and its yards set the year’s priorities
Dec 8-11, 2026	Department of War Maintenance Symposium 2026, Fort Lauderdale, FL (SAE for DoW)	**Open; rates rise after Nov 7**	Maintenance execution: the department-wide depot maintenance meeting
Jan 12-14, 2027	SNA National Symposium 2027, Arlington, VA (SNA)	Not yet open	Program priorities: where the surface force states its maintenance priorities
Jan 26-28, 2027	Military Additive Manufacturing Summit (MILAM), Tampa, FL (DSI)	**Early rate closes Oct 2**	Additive qualification: defense additive qualification and NAVSEA-spec adoption
Feb 16-18, 2027	WEST 2027, San Diego, CA (AFCEA/ USNI)	Not printed	Program priorities: the Pacific-side sea services conference
Mar 23-25, 2027	NSRP All Panel Meeting 2027, North Charleston, SC (NSRP)	“Coming soon”	Regional capacity: where yard R&D ideas get teamed program-wide
Apr 4-7, 2027	Sea-Air-Space 2027, National Harbor, MD (Navy League)	Not yet open	Program priorities: the largest US maritime defense exposition
Apr 11-15, 2027	AMPP Annual Conference + Expo 2027, Columbus, Ohio (AMPP)	**Presentation-only abstracts due Oct 1**	Preservation: the corrosion and coatings profession’s main event
Jun 8-10, 2027	MegaRust 2027, Hampton, VA (ASNE)	Attendee registration not yet open	Maintenance execution: the Navy corrosion event
Nov 2-4, 2027	Indo Pacific 2027 International Maritime Exposition, Sydney (AMDA Foundation)	Not published yet	Supply chain / policy engagement: the Pacific’s naval industry exposition; allied yards exhibit